

2 Theory of Relativity

In this section, we update our concepts of space and time.

2.1 Galilean Relativity

The Galilean Relativity states *the laws of motion are same in all inertial frames*. To be specific, the Galilean relativity refers to the invariance of Newton's law of motion. Consider two reference frames S and S' . The location of a moving particle in the two frames are (x, y, z, t) and $(x', y', z', t' = t)$ where it is assumed that the time is same in both frames. Assuming the frames are aligned, and the motion of the particle is along the X axis the Galilean transformation relations are

$$x' = x - vt, \quad y' = y, \quad z' = z, \quad t' = t.$$

2.2 Special Theory of Relativity (STR)

Maxwell's theory of light (1861) asserts that the speed of light in a vacuum is constant and does not depend on any particular frame of reference. Specifically, Maxwell's equations suggest that the speed of light is the same in all inertial frames of reference, regardless of the relative velocity between the observer and the light source. This is in direct conflict with the Galilean relativity, which assumes that velocities simply add or subtract based on the relative motion between observers. According to Galilean relativity, if a moving observer were to measure the speed of light, the result should depend on the observer's motion relative to the light source. Therefore, the fact that Maxwell's equations predict a constant speed of light contradicts Galilean relativity.

This contradiction led to a crucial question: is Maxwell's theory of light incorrect, or are Galilean transformations not the correct way to transform between inertial frames of reference?

In 1887, the Michelson-Morley experiment attempted to detect the Earth's motion through the hypothetical "ether" (a medium believed to carry light waves). They found that the speed of light in inertial frames is indeed independent of the motion of the observer relative to the supposed ether. This result suggested that the speed of light is constant in all inertial frames, further discrediting the ether hypothesis and reinforcing the implications of Maxwell's theory.

Building on the Michelson-Morley results, Einstein proposed the Special Theory of Relativity (STR) between 1902 and 1905. His theory aimed to reconcile these experimental findings with the existing framework of physics. Einstein's Special Theory of Relativity is built upon two foundational postulates:

- *The laws of physics are the same (invariant) in all inertial frames of reference.*
- *The speed of light in vacuum is constant (denoted as c) and independent of the relative motion of the source and the observer.*

Einstein's goal was to find a set of transformation rules—now known as Lorentz transformations—that would allow the description of physical phenomena from one inertial frame to another in a way that satisfies these two postulates. These transformations had already been derived by Hendrik Lorentz as part of his work on the nature of the ether, but Einstein recognized that they provided a deeper insight into the relationship between space, time, and motion.

In conclusion, Einstein's work in Special Relativity not only resolved the conflict between Maxwell's theory and Galilean relativity but also fundamentally reshaped our understanding of space and time, leading to profound insights in modern physics.

2.2.1 Lorentz Transformations

With reference to the figure [8](#), consider two reference frames, $S \equiv (x, y, z)$ and $S' \equiv (x', y', z')$, where the axes of the two frames are parallel. Assume that at $t = 0$, the origins of both frames coincide. At a later time, the relative velocity between the frames is v along the x -axis. In the S -frame, an event occurs at time t and location (x, y, z) . In the S' -frame, the same event is observed at time t' and location (x', y', z') .

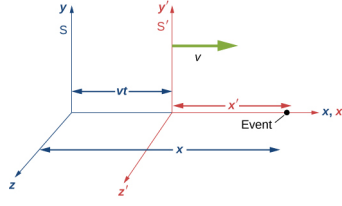


Figure 8: Two inertial frames with relative speed v .

The Lorentz transformation (LT) relations between the space and time coordinates are given by:

$$\begin{aligned} t' &= \gamma \left(t - \frac{v}{c} \frac{x}{c} \right) = \gamma \left(t - \beta \frac{x}{c} \right), \\ x' &= \gamma (x - vt) = \gamma (x - \beta ct), \\ y' &= y, \quad z' = z, \end{aligned} \tag{4}$$

where

$$\beta = \frac{v}{c}, \quad \gamma = \frac{1}{\sqrt{1 - \beta^2}}.$$

An important point to note is that when the relative velocity between the two frames is negligible compared to the speed of light (i.e., when $v \ll c$), the transformation relations simplify to:

$$t' = t, \quad x' = x - vt, \quad y' = y, \quad z' = z.$$

These are precisely the Galilean transformation equations. This implies that for everyday velocities, which are much smaller than the speed of light, the Galilean transformations provide an accurate description of the relationship between reference frames.

The set of equations above allows you to transform coordinates from frame S to frame S' . The reverse transformation, which transforms coordinates from S' to S , is obtained by simply reversing the sign of the velocity, from v to $-v$. The reverse Lorentz transformation is given by:

$$\begin{aligned} t &= \gamma \left(t' + \frac{v}{c} \frac{x'}{c} \right) = \gamma \left(t' + \beta \frac{x'}{c} \right), \\ x &= \gamma (x' + vt') = \gamma (x' + \beta ct'), \\ y &= y', \quad z = z'. \end{aligned}$$

2.2.2 Relativistic Velocity Addition

Here, we demonstrate that if the set of equations (4) represent the transformation relations between two frames, then the speed of light remains constant, regardless of the relative velocity between the light source and the observer. The proof is as follows: Consider a particle moving along the X -direction (refer to figure 8). The velocities observed in the S - and S' -frames are given by:

$$u_x = \frac{dx}{dt}, \quad u'_x = \frac{dx'}{dt'}.$$

Using the Lorentz transformation relations, we can write:

$$dx = \gamma (dx' + \beta c dt'), \quad dt = \gamma \left(dt' + \frac{\beta}{c} dx' \right).$$

Thus, the velocity in the S -frame is:

$$u_x = \frac{dx}{dt} = \frac{dx' + \beta c dt'}{dt' + \frac{\beta}{c} dx'} = \frac{u_{x'} + v}{1 + \frac{v}{c^2} u_{x'}}.$$

This is the relativistic velocity addition formula when the relative velocity between the two frames is significant. If the relative velocity is small, i.e., $v \ll c$, then the velocity in the S -frame reduces to the Galilean form:

$$u_x = u'_x + v.$$

Now, assume that the particle in question is light, so its velocity in the S' -frame is $u'_x = c$. Substituting this into the relativistic velocity addition formula gives:

$$u_x = c.$$

Therefore, the speed of light remains the same in all inertial frames, regardless of their relative velocities.

2.3 Length Contraction

In this section, we discuss an interesting consequence of Lorentz transformation (LT). With reference to figure 9, consider two reference frames S and S' , where their axes are aligned, and S' moves with velocity v relative to S along the positive x -direction. Suppose there is a stick lying along the x' -axis in the S' -frame. The observer in the S -frame measures the endpoints of the stick at positions x_1 and x_2 . The length measured in the S -frame is:

$$L = x_2 - x_1.$$

Suppose the observer in the S -frame also knows the velocity with which S' is moving. Using this information, along with his measurements, he can predict the actual length of the stick. By "actual length," we mean the length that the observer in the S' -frame would measure, which is the rest length of the stick. It can be computed as follows:

$$x'_1 = \gamma(x_1 - vt), \quad x'_2 = \gamma(x_2 - vt),$$

where the same time t is used in both equations, because the measurements in the S -frame must be simultaneous, as the object is in motion. The length measured in the S' -frame is:

$$L_0 = x'_2 - x'_1 = \gamma(x_2 - x_1) = \gamma L.$$

Rearranging this, we get the relation between the lengths:

$$L = \frac{L_0}{\gamma} = \sqrt{1 - \beta^2} L_0.$$

Thus, $L < L_0$, where L_0 is the proper length of the object — the length measured in a frame where the object is at rest. Therefore, the length of a moving object appears contracted when measured from a frame in which it is in motion.

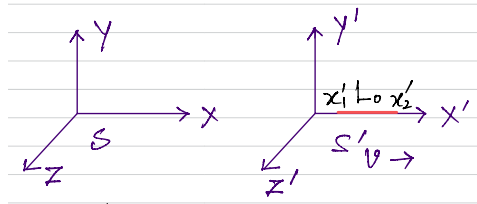


Figure 9: Two inertial frames with relative speed v .

2.4 Time Dilation

Now consider a clock located at position x' in the S' -frame. We will show that an observer in the S -frame perceives the clock to be running slower. In the S' -frame, two consecutive ticks of the clock occur at

times t'_1 and t'_2 . Using the Lorentz transformation to convert from the S' -frame to the S -frame, the times observed by an observer in the S -frame can be written as:

$$t_1 = \frac{t'_1 + \frac{v}{c^2}x'}{\sqrt{1 - \frac{v^2}{c^2}}}, \quad t_2 = \frac{t'_2 + \frac{v}{c^2}x'}{\sqrt{1 - \frac{v^2}{c^2}}}.$$

Thus, the time difference between the two ticks as observed in the S -frame is:

$$t_2 - t_1 = \frac{t'_2 - t'_1}{\sqrt{1 - \frac{v^2}{c^2}}}.$$

This shows that the time interval between the ticks is longer in the S -frame than in the S' -frame, where the clock is at rest. Hence, to an observer in the S -frame, the moving clock appears to be running slower.

This phenomenon is known as **time dilation**, which leads us to the concept of **proper time**. Proper time is defined as the time measured by an observer whose clock is at rest relative to them.

2.5 Spacetime

For convenience, we introduce the following notations:

$$x^0 \equiv ct, \quad x^1 \equiv x, \quad x^2 \equiv y, \quad x^3 \equiv z,$$

and similarly for the primed coordinates x'^1, x'^2, x'^3, x'^4 . The Lorentz transformation (LT) relations can then be written as

$$\begin{pmatrix} x'^0 \\ x'^1 \\ x'^2 \\ x'^3 \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}.$$

One crucial observation here is that the space and time coordinates are no longer separate entities; they transform together as a single entity. This combined entity is referred to as *spacetime*. Analogous to the location of a point in three-dimensional space, let us introduce the concept of an *event*, defined as a set of four coordinates (x^0, x^1, x^2, x^3) in the four-dimensional spacetime. The coordinates (x^0, x^1, x^2, x^3) define an *event* in 4D spacetime, representing the location of an event in spacetime. This concept is analogous to specifying the location of a point in 3D space. The event is often denoted by x^μ , where

$$x^\mu = (x^0, x^1, x^2, x^3) = (x^0, \vec{x}),$$

with $\mu = 0, 1, 2, 3$ being the Lorentz index, and $\vec{x}$ representing the usual three-vector.

2.5.1 Minkowski metric

The distance between two nearby points, (x, y, z) and $(x + dx, y + dy, z + dz)$, in a three-dimensional Cartesian coordinate system is given by:

$$\sqrt{dx^2 + dy^2 + dz^2}.$$

This distance remains the same (or invariant) when measured in a different coordinate system that is either translated or rotated about any axis. This invariance motivates us to look for a similar expression for the *distance between two events* in spacetime, represented by (x^0, x^1, x^2, x^3) and $(x^0 + dx^0, x^1 + dx^1, x^2 + dx^2, x^3 + dx^3)$, which remains *invariant* under Lorentz transformations.

Consider the quantity

$$ds^2 = -(dx^0)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2,$$

which represents the interval between two events in the spacetime coordinate system S . (Refer to the previous figure for a visualization.) Now suppose we perform a Lorentz transformation to a new coordinate

system S' , where the events are denoted by x'^μ and $x'^\mu + dx'^\mu$. The interval ds^2 transforms to $(ds')^2$, given by:

$$ds'^2 = -(dx'^0)^2 + (dx'^1)^2 + (dx'^2)^2 + (dx'^3)^2.$$

Using the Lorentz transformation relations, we can show (this will be done in the class) that:

$$ds'^2 = ds^2.$$

Thus, $ds^2 = -(dx^0)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2$ is an invariant quantity, called the *spacetime interval*, *line element*, or *metric*. This is analogous to the concept of length in a 3D space.

Using the summation convention, the spacetime interval can be written in a more compact form as:

$$ds^2 = -(dx^0)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2 = \eta_{\mu\nu} dx^\mu dx^\nu, \quad (5)$$

where in the last expression, we introduce the *Minkowski metric*, or simply the *metric of spacetime*, given by:

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

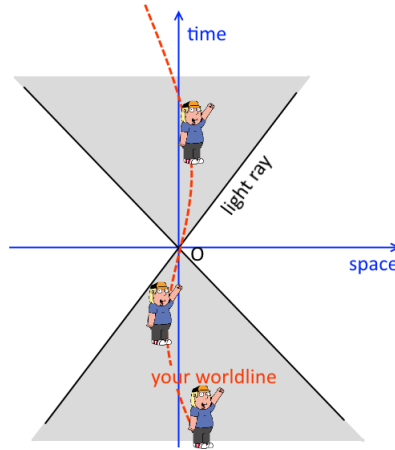


Figure 10: Light cone and your world line.

2.5.2 Light-cone

The trajectories of particles in four-dimensional spacetime are called *world lines*. The world line of light is special. Consider the line element:

$$ds^2 = -(dx^0)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2 \equiv -(dx^0)^2 + d\vec{x}^2.$$

The wavefront of light travels with velocity $c = |d\vec{x}/dx^0| = 1$ (since we set $c = 1$), meaning that the worldline of light corresponds to:

$$ds^2 = 0.$$

One can visualize the worldline of a light ray in a space vs time plot. Since it's impossible to plot this in four dimensions, we instead use a two-dimensional plot where the horizontal axis represents space and

the vertical axis represents time. The world line of a light ray forms a 45° angle with the space axis, as shown in Fig. 10. This is referred to as the *lightlike world line*. In four dimensions, this would actually be a cone, so Fig. 10 is called the *light cone*. For ordinary particles, $|dx/dt| < 1$, and thus, the worldlines of any massive particle make a larger angle with the space axis.

Due to the relative negative sign between the time and space components in the spacetime interval, ds^2 in Eq. (5) can be classified into three types:

- *Timelike* ($ds^2 < 0$): Two events separated by $ds^2 < 0$ are causally connected, meaning that information can travel from one event to the other. An example is two events occurring at the same spatial location but at different times.
- *Spacelike* ($ds^2 > 0$): Spacelike events are causally disconnected. For instance, two events that are simultaneous in time but separated in space.
- *Null or lightlike* ($ds^2 = 0$): These events are causally connected and lie along the world line of light.